

When Does Aggregating Explanations Work?

RetinaMNIST Results

Jinhua Xu, Davide Anguita, Fabio Roli, Jing Yuan, and Luca Oneto

This report consolidates the released **RetinaMNIST** experiment results. Tables are grouped by model, experimental protocol, and aggregation setting. Metric directions and robustness definitions are stated in every table header and caption.

Protocol	Model	Settings
Full-matrix protocol	ViT-B/16	NAIVE, IND, NOISE-S, NOISE-K

Table 1: NAIVE results for **RetinaMNIST** with ViT-B/16 (Full-matrix protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains $n = 400$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _F ^g $\uparrow$	R _C ^g $\uparrow$	R _C ^g $\uparrow$	R _F ^p $\uparrow$	R _F ^p $\uparrow$	R _C ^p $\uparrow$	R _C ^p $\uparrow$	R _F ^s $\uparrow$	R _F ^s $\uparrow$	R _C ^s $\uparrow$	R _C ^s $\uparrow$	R _F ^a $\uparrow$	R _F ^a $\uparrow$	R _C ^a $\uparrow$	R _C ^a $\uparrow$
Best Individual	0.083	0.005	0.422	0.365	-0.175	0.235	0.200	-0.370	-0.110	0.092	0.095	-0.190	-0.060	0.010	0.147	-0.160	-0.008	0.005	-0.013	0.018
Simple Averaging	0.015	0.003	0.370	0.432	-0.050	0.195	0.302	-0.282	-0.015	0.088	0.195	-0.100	0.048	0.020	0.205	-0.093	0.015	-0.035	-0.013	0.008
Borda	0.025	0.052	0.380	0.477	-0.083	0.247	0.247	-0.225	-0.013	0.122	0.132	-0.038	0.043	0.048	0.167	-0.053	0.033	0.005	-0.033	0.070
Kemeny–Young	0.065	0.040	0.357	0.472	-0.130	0.258	0.347	-0.242	-0.060	0.103	0.217	-0.040	-0.025	0.020	0.163	-0.055	-0.013	0.007	-0.040	0.057
RRF	0.040	0.043	0.375	0.468	-0.122	0.255	0.300	-0.247	-0.033	0.100	0.170	-0.055	0.037	0.013	0.172	-0.060	0.005	-0.005	-0.050	0.048
Schulze	0.058	0.037	0.370	0.468	-0.130	0.240	0.300	-0.257	-0.040	0.103	0.182	-0.055	0.000	0.022	0.162	-0.057	-0.003	-0.010	-0.030	0.055

Table 2: IND results for **RetinaMNIST** with ViT-B/16 (Full-matrix protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at $q = 11$. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains $n = 400$ samples.

Method	Setting	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _F ^g $\uparrow$	R _C ^g $\uparrow$	R _C ^g $\uparrow$	R _F ^p $\uparrow$	R _F ^p $\uparrow$	R _C ^p $\uparrow$	R _C ^p $\uparrow$	R _F ^s $\uparrow$	R _F ^s $\uparrow$	R _C ^s $\uparrow$	R _C ^s $\uparrow$	R _F ^a $\uparrow$	R _F ^a $\uparrow$	R _C ^a $\uparrow$	R _C ^a $\uparrow$
Best Individual	baseline	-0.011	0.105	0.199	0.489	-0.063	0.288	0.217	-0.275	-0.012	0.163	0.194	-0.133	-0.023	0.060	0.085	-0.121	0.009	0.017	-0.027	0.023
Simple Averaging	matched NAIVE	-0.023	0.127	0.163	0.534	-0.061	0.292	0.233	-0.224	-0.023	0.157	0.184	-0.066	-0.012	0.076	0.108	-0.077	0.022	0.013	-0.005	0.026
Simple Averaging	IND	-0.018	0.120	0.188	0.522	-0.048	0.333	0.285	-0.225	0.033	0.185	0.232	-0.035	0.008	0.110	0.135	-0.033	0.018	0.022	0.003	0.032
Borda	matched NAIVE	-0.025	0.122	0.167	0.532	-0.069	0.282	0.220	-0.242	-0.018	0.170	0.182	-0.059	-0.021	0.066	0.115	-0.078	0.019	0.012	0.001	0.008
Borda	IND	-0.022	0.150	0.212	0.532	-0.078	0.367	0.207	-0.205	-0.028	0.198	0.145	-0.023	-0.005	0.120	0.100	-0.030	0.025	-0.003	-0.038	-0.003
Kemeny–Young	matched NAIVE	-0.020	0.147	0.152	0.557	-0.065	0.332	0.257	-0.216	-0.028	0.188	0.189	-0.046	-0.012	0.094	0.126	-0.046	0.017	0.033	-0.005	0.032
Kemeny–Young	IND	-0.035	0.128	0.235	0.537	-0.045	0.348	0.237	-0.215	0.040	0.182	0.163	-0.040	0.020	0.113	0.093	-0.003	0.028	0.000	-0.055	0.032
RRF	matched NAIVE	-0.012	0.127	0.154	0.557	-0.069	0.296	0.236	-0.223	-0.034	0.183	0.186	-0.044	0.002	0.068	0.107	-0.053	0.012	0.016	-0.007	0.020
RRF	IND	-0.020	0.090	0.228	0.502	-0.067	0.305	0.255	-0.250	0.022	0.170	0.187	-0.068	0.030	0.090	0.103	-0.045	0.018	-0.018	-0.048	0.002
Schulze	matched NAIVE	-0.022	0.119	0.164	0.541	-0.077	0.283	0.239	-0.229	-0.022	0.163	0.186	-0.059	-0.023	0.078	0.121	-0.064	0.018	0.010	0.001	0.034
Schulze	IND	-0.022	0.117	0.235	0.527	-0.062	0.333	0.218	-0.228	-0.025	0.182	0.150	-0.020	0.005	0.105	0.110	-0.025	0.005	-0.020	-0.067	0.017

Table 3: NOISE-S results for **RetinaMNIST** with ViT-B/16 (Full-matrix protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _F ^g $\uparrow$	R _C ^g $\uparrow$	R _C ^g $\uparrow$	R _F ^p $\uparrow$	R _F ^p $\uparrow$	R _C ^p $\uparrow$	R _C ^p $\uparrow$	R _F ^s $\uparrow$	R _F ^s $\uparrow$	R _C ^s $\uparrow$	R _C ^s $\uparrow$	R _F ^a $\uparrow$	R _F ^a $\uparrow$	R _C ^a $\uparrow$	R _C ^a $\uparrow$
Simple Averaging	3	0.077	0.130	0.287	0.540	-0.152	0.330	0.358	-0.195	-0.100	0.198	0.233	0.020	-0.018	0.118	0.210	0.008	-0.018	0.005	-0.042	0.035
Borda	3	0.070	0.125	0.287	0.535	-0.140	0.312	0.368	-0.210	-0.088	0.188	0.235	-0.002	-0.018	0.113	0.180	0.015	-0.018	0.010	-0.052	0.033
Kemeny–Young	3	0.083	0.182	0.282	0.595	-0.165	0.372	0.338	-0.150	-0.103	0.255	0.248	0.055	-0.050	0.160	0.188	0.050	-0.018	0.045	-0.040	0.073
RRF	3	0.072	0.133	0.285	0.542	-0.155	0.323	0.347	-0.198	-0.077	0.193	0.250	0.010	-0.010	0.120	0.183	0.012	-0.022	0.023	-0.052	0.047
Schulze	3	0.083	0.142	0.287	0.555	-0.160	0.335	0.353	-0.182	-0.095	0.217	0.243	0.028	-0.043	0.130	0.193	0.030	-0.018	0.015	-0.047	0.040

Table 4: NOISE-K results for **RetinaMNIST** with ViT-B/16 (Full-matrix protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Simple Averaging	9	0.003	0.005	0.378	0.438	-0.058	0.198	0.295	-0.277	-0.005	0.083	0.197	-0.080	0.065	0.015	0.197	-0.088	0.042	-0.017	-0.010	0.037
Borda	9	0.048	0.035	0.365	0.465	-0.128	0.238	0.277	-0.253	-0.052	0.115	0.160	-0.050	0.025	0.038	0.200	-0.037	0.007	-0.005	-0.003	0.068
Kemeny–Young	9	0.052	0.048	0.347	0.500	-0.130	0.258	0.347	-0.233	-0.037	0.113	0.225	-0.030	0.007	0.058	0.170	-0.022	0.003	-0.005	-0.042	0.062
RRF	9	0.055	0.045	0.375	0.480	-0.140	0.258	0.305	-0.240	-0.028	0.120	0.177	-0.042	0.022	0.035	0.142	-0.032	-0.010	0.000	-0.055	0.068
Schulze	9	0.060	0.045	0.355	0.465	-0.135	0.250	0.320	-0.255	-0.040	0.110	0.198	-0.040	0.003	0.048	0.175	-0.030	-0.007	-0.013	-0.037	0.040